LEARNER GUIDE

The tangent ratio

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

In Lesson 1 you saw that a fixed angle keeps the ratio of two sides constant; in Lesson 2 you gave the sides their names. This lesson brings those together and gives the first ratio its own name. For a marked angle θ in a right-angled triangle, the ratio opposite ÷ adjacent is called the tangent of the angle, written tan θ. Because the ratio is the same for every triangle with that angle, tan θ is a single number that belongs to the angle. You watch it change as the angle changes — a bigger angle makes a steeper triangle and a bigger tangent — and you use the tangent to compare how steep different slopes are without measuring a single angle. The question you are exploring is: how does opposite ÷ adjacent capture an angle?

What you are learning to notice

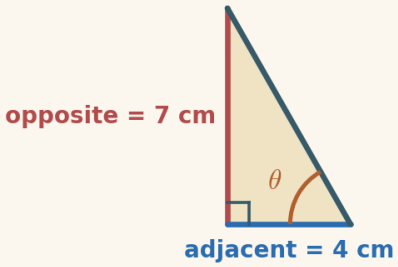
- That tan θ = opposite ÷ adjacent — the opposite side divided by the adjacent side, never the hypotenuse.
- That tan θ is the same for every right-angled triangle with that angle, whatever its size.
- That as the angle gets bigger the triangle gets steeper and tan θ gets bigger.
- That the tangent lets you compare and order slopes by steepness using a single number.

Moving through it, one step at a time

In The ratio with a name, work out opposite ÷ adjacent for three triangles of the same shape and see they all give the same number — that number is tan θ. In Tangent Explorer, drag the angle slider: the adjacent side stays the same while the opposite side and the value of tan θ grow with the angle. In Is this tan θ?, judge claimed values as right or a slip, watching for the two classic errors — using the hypotenuse, and turning the ratio upside down. In Rank by steepness, work out tan θ for three slopes given as rise over run and order them from gentlest to steepest. Use the Scratchpad to draw a triangle, mark θ and work out tan θ for yourself.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.



Example 1 — finding the tangent

The tangent of the marked angle is the opposite divided by the adjacent. Here the opposite is 7 cm and the adjacent is 4 cm.

1. the tangent uses opposite and adjacent	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$
2. put in the numbers	$\tan\theta = \frac{7}{4}$
3. work it out	$\tan\theta = 1.75$

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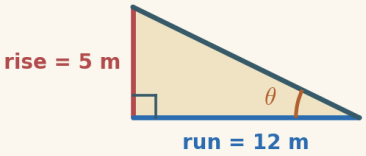

Example 2 — the tangent of a ramp

A ramp rises 5 m over a run of 12 m. Its rise is the opposite and its run is the adjacent, so its tangent is rise ÷ run.

1. the tangent uses opposite (rise) and adjacent (run)	$\tan\theta = \frac{\text{rise}}{\text{run}}$
2. put in the numbers	$\tan\theta = \frac{5}{12}$
3. work it out	$\tan\theta \approx 0.42$

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A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (naming the opposite and adjacent sides relative to a marked angle (Lesson 2), the constant side ratio for a fixed angle (Lesson 1), and a ratio written as a division (a ÷ b)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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